

Assignment 2: A Critical Review of Ateeq et al. (2024),

Artificial intelligence in education:

Implications for academic integrity and the shift toward holistic assessment

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Study:

Ateeq, A., Alzoraiki, M., Milhem, M., & Ateeq, R. A. (2024). Artificial intelligence in education: Implications for academic integrity and the shift toward holistic assessment. *Frontiers in Education*, 9, 1470979. <https://doi.org/10.3389/feduc.2024.1470979>

Introduction:

With the arrival of Artificial Intelligence, many teachers are wondering about how this will impact their practice. Some are hopeful to see improvements in academic outcomes, while others are afraid of a growing inability of students to think critically for themselves. How are schools going to handle the rapid integration of generative tools like ChatGPT and Gemini? This critique examines the study by Ateeq et al. (2024), “Artificial intelligence in education: implications for academic integrity and the shift toward holistic assessment.” It claims to be particularly focussed on the issue of academic integrity, which is a legitimate concern, but the research questions of the study and the survey tool seem to be misaligned.

There are three research questions and three corresponding research objectives. The first is about the long-term effects of AI-cheating in online learning and on credential value. The second examines the impact of application-centric assessments versus knowledge-based assessments. The third asks how educational policy can affect AI-cheating. The authors of the study conclude that AI in education raises integrity concerns, but well-designed education with AI, along with policy, ethics and pedagogy, most strongly predicts academic success.

Instrumentation:

The study used a Likert-scale survey of 22 questions, organized around four constructs: six questions each on Educational Impact (EI) and Policy and Ethics (PE), and five questions each on Pedagogical Implications (PI) and Academic Outcomes (AO).

The study relies on external literature to establish the existence and severity of cheating with AI. The survey does not measure the existence of cheating. What the study does analyze is the influence of three of the constructs (independent variables EI, PE, and PI) on the dependent variable of Academic Outcomes (AO).

The study used a quantitative, cross-sectional survey design. The sample consisted of 203 participants, comprising both university students and faculty members in Bahrain, selected via convenience sampling from an initial pool of 250 invitees. Data was analyzed using Partial Least Squares Structural Equation Modelling (PLS-SEM) to see how well their constructs predicted Academic Outcomes. SPSS (Statistical package for the social sciences) was used only for descriptive statistics (means and standard deviations) and for calculating Cronbach's alpha to assess internal consistency for each construct; it did not generate the main hypothesis-testing findings. AMOS (Analysis of Moment Structures) was mentioned but used minimally.

A primary strength of this study is its timeliness and relevance. In an educational landscape currently dominated by "panic" over AI plagiarism, Ateeq et al. (2024) move the conversation beyond the binary of banning vs. embracing AI. Instead, they offer what they claim to be a data-driven validation for "holistic assessment," arguing that assessments requiring critical thinking and creativity are naturally more resistant to AI misuse.

Statistically, the study demonstrates rigour. The researchers established high internal consistency for their survey instruments, with Cronbach's alpha values exceeding the 0.7 threshold for all constructs.

In the results section, the authors explain why the measurement instrument of their study should be seen as trustworthy by highlighting the statistical quality of their study:

The table presents the Cronbach's alpha (α), composite reliability (CR), and average variance extracted (AVE) of four constructs: AO, EI, PE, and PI. Firstly, all constructs show an α greater than 0.7, suggesting good internal consistency ([Abro and Salam, 2014](#); [Milhem et al., 2019](#)). In terms of CR, all values exceed the 0.7 threshold, indicating satisfactory reliability, with PI having the highest CR at 0.944 ([Al-Fahim et al., 2024](#)).¹ Regarding Average Variance Extracted (AVE), only Average Obstruction (AO)² falls below the recommended threshold of 0.5, indicating weak convergent validity.³ Conversely, the values of PI, EI, and PE exceed the established threshold, likely indicating the presence of issues ([Milhem et al., 2024](#); [Milhem et al., 2024](#)).⁴ In conclusion, although all variables demonstrate strong reliability, there is need for improvement in the convergent validity of (AO), (EI), (PE).⁵ (Ateeq et al., 2024, p. 5)

However, the authors seem to know that some aspects are less than ideal. The study itself states that the R-square value for AO is 0.389 and admits that although the model has “some explanatory power, a significant portion of the variance remains unexplained” (Ateeq et al., 2024). The less than stellar AVE for AO input most likely contributed to the R^2 output. There is cause to mistrust the latent variable (the AO construct) since the survey items are problematic.

Theoretically, the study aligns well with modern progressive pedagogies, such as Universal Design for Learning (UDL) and project-based learning, but there are no empirically validating data for this. None of the survey questions mention “application-centric” assessment of any kind, even though one of the research objectives is “Evaluate the impact of application-centric assessments versus traditional knowledge-based assessments on student understanding, creativity, and critical thinking”

(Ateeq et al., 2024). The study cites other sources and simply says that research supports the idea that flexible, multimodal ways of demonstrating knowledge are superior to rigid, standardized testing.

Critical Analysis: Weaknesses and Limitations

It is important that quantitative studies demonstrate rigour and control bias (Suter, 2012). The main problem in the Ateeq et al. (2024) study is conceptual misalignment between the research questions and the measurement instrument. The stated research questions focus on AI-cheating, long-term credential impacts, and the influence of educational policy, but the survey only collects participants' beliefs about pedagogy, ethics, and perceived academic outcomes. The study cannot empirically address several of its own research questions because it does not collect data on cheating and credential impacts. At the very least, there should have been survey questions like, "How often do you use AI to draft assignment responses without citation?" The research does capture positive attitudes toward AI in education, but that cannot generate evidence about actual learning outcomes or instances of misconduct.

The use of convenience sampling introduces potential external validity issues, a threat widely discussed in methodological literature (Suter, 2012). Because participants were self-selected, the findings may not be generalizable due to possible bias. Participants who volunteered may have been those with strong opinions on AI or higher digital literacy, skewing the results. Additionally, the study is geographically limited to the Kingdom of Bahrain. Cultural attitudes toward academic authority, cheating, and technology vary significantly across regions; thus, findings from this specific context may not transfer reliably to North American, Asian, or European universities.

By capturing data at a single point in time, the cross-sectional study can demonstrate correlation but not causation (Suter, 2012). For example, it is unclear whether "Pedagogical Implications" caused better "Academic Outcomes," or if high-performing students simply adapted better to new pedagogical

tools. The positive results themselves must be scrutinized since the study relies on self-reported data. Faculty members may have over-reported their use of ethical teaching practices (Social Desirability Bias), while students may have under-reported their reliance on AI tools to appear more academically honest. The Likert-scale survey items are all positive statements about AI use (e.g., “AI can...”, “AI will...”, “AI enhances...”; there are no reverse-worded items). There is a risk that a respondent might tend to agree to the majority of items due to Acquiescence bias (Ilhan et al., 2024). Statistically speaking, this allows for stronger internal consistency but decreases confidence in external validity.

Apart from these issues, the study includes concerning inconsistencies; a surprising sloppiness that cannot be ignored. As well as mislabeling AO as “average obstructions”, the beta-coefficient described in the abstract for PI ($\beta=0.454$) does not match the data in the published tables. I have assumed the much lower figure in the data table to be the correct one, but this further undermines the trustworthiness of the study.

Implications for Professional Practice

The findings of Ateeq et al. (2024) have useful implications for professional practice, particularly in the shift from “policing” to “designing.” The beta-coefficients (which are not causal in a cross-sectional study; Suter, 2012) reported that Pedagogical Implications ($\beta=0.218$) have a measurably stronger influence than Policy and Ethics ($\beta=0.156$) in driving academic outcomes. For educators, this is a helpful directive: investing energy in designing better assignments is potentially more effective than investing energy in enforcing stricter bans.

In the context of fostering student agency, this supports a move away from formulaic writing templates towards multimodal composition. If “application-centric” assessment is the goal, then students should be encouraged to create artifacts (compositions, design projects, events, and experiences) that demonstrate understanding. This approach not only upholds academic integrity (as

these tasks are harder to fake) but also empowers students to use AI as a collaborative tool for creativity rather than a shortcut for rote work.

Writing this critique has inspired some new teaching goals as a high school technology and music teacher. I want to incorporate more opportunities for students to exercise their learning agency and create useful artifacts with and without AI. I hope to research the benefits using Action Research by developing authentic measurement tools that can assess the effectiveness of using AI as a co-creation tool.

Conclusion

Ateeq et al. (2024) provide a valuable starting point for thinking about AI's disruptive impact on higher education and helps to fill the gap between problems with AI use in academia and where to go from here. While the study's generalizability is limited by its sampling method and geographical focus, its core message – that academic integrity is better supported through engagement and thoughtful assessment design than through surveillance – is compelling. However, I was hoping the study would be more convincing methodologically. The measurement instrument is statistically robust, but the research questions are not fully addressed by the items included in the survey, creating a shortfall between what the authors claim to investigate and what the study actually measures. As a result, the findings describe a direction I agree with and hope to see continue, but they cannot serve as definitive proof. The study's design ultimately fails in providing the level of evidence needed to substantiate its broader claims, even though its argument for holistic, application-centric assessment remains pedagogically meaningful.

References

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Footnotes

¹ Reliability Confirmation: The reporting of Cronbach's alpha ($\alpha > 0.7$) and Composite Reliability (CR > 0.7) for all constructs indicates satisfactory internal consistency. This strengthens the measurement model by demonstrating that the survey items reliably measure their intended latent variables.

² The authors likely meant "Academic Outcomes" instead of "Average Obstructions."

³ Methodological Weakness: The AO construct fails to meet the AVE threshold of 0.5, indicating weak convergent validity. This is a critical limitation, as it suggests the items designed to measure AO do not adequately represent the construct, potentially undermining the accuracy of any conclusions tied to this variable.

⁴ Connecting measurement quality to method (PLS-SEM): The detailed reporting of α , CR, and AVE is essential for evaluating the measurement model in PLS-SEM. This methodology requires assessment of the outer model (measurement model) before proceeding to the inner model (hypothesis testing). It is not clear whether the high AVE numbers point to a problem or if this was a misstatement.

⁵ Critique of Interpretation: The low AVE for AO is a concern, but high AVE values for PI, EI, and PE (above 0.55) typically indicate satisfactory convergent validity. The authors' particular listing of constructs is confusing here.

AI use

I used ChatGPT to check spelling and grammar. NotebookLM was used to organize resources. ChatGPT, Gemini, and NotebookLM were used to help me understand certain statistical symbols and how statistical software worked. ChatGPT was used to understand which statistical data was produced by SPSS and which came from PLS-SEM. Some of the sentence phrasing was shaped and adjusted by AI, but the ideas are my own.

Summary and Use of Peer Feedback

I received valuable and encouraging feedback from two of my peers. I feel I was able to improve my critique by incorporating the following suggestions: 1) provide an example of a “missing construct” question regarding cheating in the survey instrument; 2) tightening some of the statistical analysis around weak convergent validity; and 3) to ground the “implications” section in my specific context as a high school teacher.